

AI-ML FOR MECHANICAL ENGINEERS

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Outline

1 SESSION 1: AI OVERVIEW

Introduction to AI

Data Science, Artificial Intelligence is critical in bringing intelligent automation

What are Data Sciences?

ie What is Artificial Intelligence? Machine Learning? Deep Learning?

Data Science

- ▶ Science of Data (obviously)
- ▶ Use of Data for Applications
- ▶ Some parts of AI uses Data to find patterns and insights which are helpful in multiple applications
- ▶ Machine and Deep Learning that part of AI that leverages data.

So, more on AI-ML here ...

What is the Core Idea?

What's the core idea?

- ▶ behind problem solving?
- ▶ behind writing software algorithms?
- ▶ solving research problems?

Desire

- ▶ To find a “function”
- ▶ To find a relation
- ▶ To find a transformation
- ▶ To build a model
- ▶ From given inputs to desired outputs.

That's it.

Functions

- ▶ Some functions are straight forward
- ▶ *"In summer, ice-cream sale goes up"*
- ▶ Cause and effect
- ▶ Relation (function, Mathematical model) is found out
- ▶ Here, simple rule based programming suffices

Functions

- ▶ But some functions are complex
- ▶ *"More you put efforts, your business flourishes."*
- ▶ Cause and effect again, but the relation is far to complex
- ▶ Too many variables
- ▶ Here, simple rule based programming not humanly possible.
- ▶ Lots of research needed to come up with equations.

Functions

- ▶ $E = mc^2$
- ▶ What's this? a function?
- ▶ Input variable(s)?
- ▶ Output variable(s)?
- ▶ Parameters?
- ▶ How's the relation? linear?

Functions

- ▶ But most real-life functions are not deterministic
- ▶ Some are probabilistic, some non-linear.
- ▶ *“Detecting if the tumor is benign or malignant”*
- ▶ *“At any state in the game of chess, whats the next move?”*

Chess: next move?

- ▶ Needs extreme expertise
- ▶ Needs “intelligence”
- ▶ How do you get that?
 - ▶ Built by lots of training.
 - ▶ By studying lots of past games.
- ▶ This is how Humans build intelligence

Intelligence

- ▶ Can machine (software/program) also do the same?
- ▶ Can it play chess?
- ▶ Can it build intelligence?
- ▶ By looking at past experiences (data),
- ▶ Training Data: games played, moves used, etc.

Yes, it can!! Thats Artificial Intelligence.

What is AI?

What is Artificial Intelligence (AI)?

My definition:

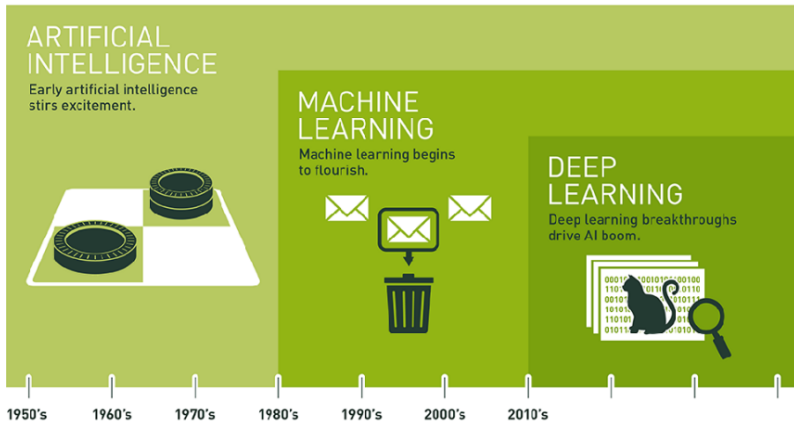
“If machines (or computer programs) start doing some/all of these “intelligent” tasks, then that’s Artificial Intelligence”

Intelligence: the differentiation

- ▶ Ability to think various domains
- ▶ Ability to produce something new
- ▶ Ability to detect the unseen
- ▶ Ability to enhance knowledge (rules, patterns)

All these, AI has started doing. The AI era has arrived!!

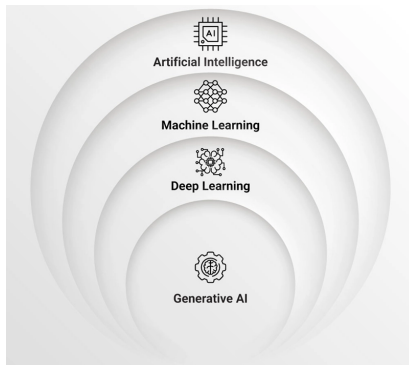
Relationship between AI, ML, DL



(Ref: <https://blogs.nvidia.com/blog/2016/07/29/whats-difference-artificial-intelligence-machine-learning-deep-learning-ai/>)

The Modern AI Hierarchy

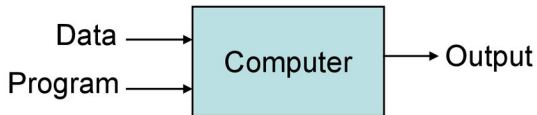
- ▶ **Artificial Intelligence (AI):** Machines mimicking human intelligence.
- ▶ **Machine Learning (ML):** Learning from data without explicit programming.
- ▶ **Deep Learning (DL):** Neural networks with many layers.
- ▶ **Generative AI (GenAI):** DL models that can *create* new content (text, code, images) rather than just analyzing existing data.



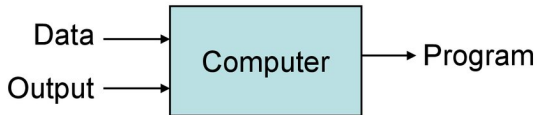
(Ref: AI, ML, DL, and Generative AI Face Off: A Comparative Analysis - Synoptek)

Traditional vs. Machine Learning?

Traditional Programming



Machine Learning



Types of Machine Learning

- ▶ Supervised Learning (with labeled data example: spam detection)
- ▶ Unsupervised Learning (clustering example: customer segmentation)
- ▶ Reinforcement Learning (gaming example: AlphaGo)

Why Machine/Deep Learning?

- ▶ Problems with High Dimensionality
- ▶ Hard/Expensive to program manually
- ▶ Techniques to model 'ANY' function given 'ENOUGH' data.
- ▶ Job \$\$\$

Why now?

- ▶ Flood of data (Internet, IoT)
- ▶ Increasing computational power
- ▶ Easy/free availability of algorithms
- ▶ Increasing support from industries

Evolution Timeline

- ▶ 1950s: Perceptron (Rosenblatt)
- ▶ 1980s: Backpropagation popularized
- ▶ 1997: Deep Blue beats Kasparov
- ▶ 2006: Deep Learning revolution (Hinton)
- ▶ 2012: AlexNet wins ImageNet
- ▶ 2016: AlphaGo beats Lee Sedol
- ▶ 2017: Transformers introduced (Attention is All You Need)
- ▶ 2018: BERT, GPT-1
- ▶ 2020: GPT-3
- ▶ 2022: ChatGPT launches
- ▶ 2023: GPT-4, Claude 2, Gemini 1.0
- ▶ 2024: Claude 3/3.5, multimodal models mainstream, AI coding tools
- ▶ 2025–26: Agentic AI, reasoning models (o1, o3, DeepSeek-R1), AI agents in production

The Foundation Model Revolution (2022–2025)

- ▶ **Old paradigm:** One model per task: train a spam filter, train a translator, train a chatbot separately. Each required thousands of labeled examples.
- ▶ **New paradigm:** One massive model (GPT-4, Claude, Gemini), trained on internet-scale data, handles *any* task via prompting alone.
- ▶ ChatGPT does translation, coding, writing, legal analysis, and data visualization: all from a single model.
- ▶ **Career implication:** You no longer need to *build* models. The value has shifted to knowing how to *use, connect, and customize* them for a specific domain.

(Ref: Foundation Models: Stanford HAI 2021; OpenAI technical reports 2023)

The Modern AI Model Landscape

Proprietary (API access):

- ▶ GPT-4o, o3 (OpenAI)
- ▶ Claude 3.5/4 Sonnet (Anthropic)
- ▶ Gemini 2.0 Flash (Google)

Open-weight (run locally):

- ▶ Llama 3.3 70B (Meta)
- ▶ DeepSeek-R1, Qwen 2.5
- ▶ Mistral Large 2

(Ref: LMSYS Chatbot Arena; Hugging Face Model Hub 2025)

Key insight for your career:

- ▶ Models are **commoditizing fast** — capability gaps shrink quarterly
- ▶ Your value is in *application* and *domain judgment*, not in which model you chose
- ▶ API costs drop $\sim 10\times$ every 12–18 months

Multimodality: Beyond Text

- ▶ Modern AI handles **multiple input and output types** simultaneously:
 - ▶ **Input:** Text, images, audio, video, documents, code
 - ▶ **Output:** Text, code, images, structured data, voice
- ▶ **Real examples:**
 - ▶ GPT-4o: Real-time voice conversations with emotional awareness
 - ▶ Gemini 2.0: Live video analysis (“what’s on my desk?”)
 - ▶ DALL-E 3 / Midjourney v6: Text to photorealistic images
 - ▶ Sora: Text to video clips
- ▶ **Career implication:** Domain experts who feed domain-specific images, scans, or documents to AI unlock capabilities that pure coders cannot replicate

Agents & Agentic AI (2024-2025)

- ▶ What are AI agents?
 - ▶ Autonomous systems that can plan, execute, and adapt
 - ▶ Examples: LangGraph, CrewAI, OpenAI Assistants API
- ▶ Real applications:
 - ▶ Customer service agents (resolving tickets end-to-end)
 - ▶ Research agents (gathering and synthesizing information)
 - ▶ Coding agents (Claude Code, GitHub Copilot)
- ▶ Key shift: From "ask AI" to "AI does the task"

Real-World AI Success Stories (2024)

- ▶ Healthcare: AI detecting cancer earlier than doctors (Study: Nature 2024)
- ▶ Climate: AI optimizing energy grids (Google DeepMind 15% reduction)
- ▶ Science: AlphaFold solving protein folding (Nobel Prize 2024)
- ▶ Education: Personalized tutoring at scale (Khan Academy)

Everyday usage

Artificial intelligence seems to have become ubiquitous.

- ▶ Replying to our emails on Gmail
- ▶ Learning how to drive our cars,
- ▶ Sorting our holiday photos.
- ▶ Buy recommendations
- ▶ Face unlock on phone
- ▶ Google Maps traffic predictions
- ▶ etc.

Purpose: Make AI tangible and relatable immediately
Too good to be true, isn't it, sort of Magical !!

Is AI a threat?

But then ...

- ▶ When its too good, you start suspecting
- ▶ Is it for real!!
- ▶ How can such thing happen?
- ▶ How far will it go?

The next thing you know, people are worrying about exactly how and when AI is going to doom humanity.

AI Threat? Reality vs. Fiction

Popular fears:

- ▶ Elon Musk: AI is humanity's "biggest existential threat"
- ▶ Sci-fi movies (Terminator, Ex Machina): rogue superintelligence
- ▶ AI-assisted autonomous weapons are no longer purely fiction



Current reality:

- ▶ Today's AI is *narrow* — excellent at specific tasks, not general reasoning
- ▶ Real near-term risks: misinformation, bias, job displacement — not robot uprisings

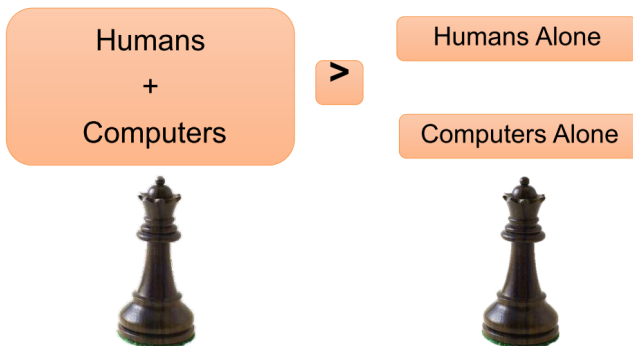
(Ref: What is Artificial Intelligence — Artificial Intelligence Tutorial For Beginners — Edureka)

Fear: Are we being replaced?

- ▶ Yes. in tasks that are repetitive
- ▶ But not which require complex thinking and creativity

Mostly

Technology Enhancing (Not Replacing) Humans



(Ref: "Artificial Intelligence Overview" - Harry Surden)

Limits on Artificial Intelligence

- ▶ Many things still beyond the realm of AI
- ▶ No thinking computers
- ▶ No Abstract Reasoning
- ▶ Often AI systems Have Accuracy Limits
- ▶ Many things difficult to capture in data
- ▶ Sometimes Hard to interpret Systems

AI Bias & Fairness

- ▶ Real examples of AI failures
 - ▶ Amazon hiring tool (2018) - biased against women
 - ▶ Facial recognition errors (higher for minorities)
 - ▶ Loan approval algorithms (redlining concerns)
- ▶ What's being done:
 - ▶ Fairness metrics
 - ▶ Diverse training data
 - ▶ Regulatory frameworks (EU AI Act)

Environmental Cost of AI

- ▶ The hidden impact
 - ▶ Training GPT-3: approx. 552 tons CO₂
 - ▶ Water usage for cooling data centers
 - ▶ Energy consumption trends
- ▶ Counter-balance:
 - ▶ AI optimizing renewable energy
 - ▶ Climate modeling improvements
- ▶ Message: "Innovation with responsibility"

AI Regulation Landscape (2024-2025)

- ▶ Global approaches:
 - ▶ EU AI Act (Risk-based regulation)
 - ▶ US Executive Order (Safety standards)
 - ▶ China's approach (State control)
 - ▶ India's stance (Innovation-first)
- ▶ Message: "Innovation with responsibility"

Jobs of the Future

- ▶ **Domain roles being transformed** (not eliminated):
 - ▶ Doctors → AI-augmented diagnosticians
 - ▶ Lawyers → Legal + AI researchers
 - ▶ Teachers → Learning experience designers
 - ▶ Engineers → AI-assisted system designers
- ▶ **Skills that remain irreplaceable:** Critical thinking, creativity, domain judgment, ethical reasoning
- ▶ **New baseline:** AI literacy is the new computer literacy — non-negotiable by 2027

(Ref: WEF Future of Jobs Report 2023; McKinsey Global Institute 2024)

AGI - The Ultimate Goal?

What is Artificial General Intelligence?

- ▶ Human-level intelligence across ALL tasks
- ▶ Current AI: Narrow/Specialized

Timeline predictions:

- ▶ Optimists: 2030s (Sam Altman, Demis Hassabis)
- ▶ Skeptics: Centuries or never (Yann LeCun)

Key challenges:

- ▶ Common sense reasoning
- ▶ Transfer learning
- ▶ Consciousness (philosophical question)

Summary

When to Use AI vs. Traditional Programming

- ▶ Clear rules? → Traditional programming
- ▶ Pattern recognition needed? → AI/ML
- ▶ Large unstructured data? → Deep Learning
- ▶ Need creativity/generation? → GenAI

Getting Started with AI - Your Action Plan

For Beginners:

- ▶ Play with ChatGPT/Claude (free tiers)
- ▶ Take: fast.ai course (free)
- ▶ Follow: Andrej Karpathy, Andrew Ng

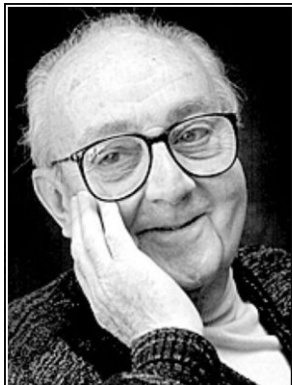
For Developers:

- ▶ Build with: Hugging Face, LangChain
- ▶ Try: Fine-tuning small models
- ▶ Contribute: Open source AI projects

For Everyone:

- ▶ Stay informed: AI newsletters
- ▶ Attend: Local AI meetups
- ▶ Think critically: Question AI claims

After all this - The Truth ...



All models are wrong, but some are useful.

— *George E. P. Box* —

AZ QUOTES